

Tian-Yi Zhou (she/her)

CONTACT INFORMATION

500 W 120th St # 535
New York, NY 10027
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RESEARCH INTERESTS

Deep learning, cybersecurity, statistical learning theory, operator learning, functional data analysis, computational social science

EDUCATION / EMPLOYMENT

- Columbia University**, New York, NY
Postdoctoral Researcher at the Data Science Institute July 2025 - July 2027
- Mentors: Samory Kpotufe (Department of Statistics), Qiang Du (Department of Applied Physics and Applied Mathematics)
- Georgia Institute of Technology**, Atlanta, GA
Ph.D. in Machine Learning, GPA: 4.00/4.00 August 2020 - May 2025
- H. Milton Stewart School of Industrial and Systems Engineering (ISyE), Minor in Operations Research
 - Thesis Advisor: Xiaoming Huo (ISyE)
- University of California, Los Angeles**, Los Angeles, CA Jan 2019 - June 2019
Exchange Student, Study Mathematics and Statistics
- University of Hong Kong**, Hong Kong SAR Sep 2016 - June 2020
B.Sc. in Statistics, First Class Honors
- Department of Statistics and Actuarial Science

PUBLICATIONS

The titles contain an embedded link to the paper.
* indicates equal distribution. α indicates alphabetical order.

1. **Drink like a Man? Modified Poisson Analysis of Adolescent Binge Drinking in the US, 1976–2022.** (2025)
Jiaxin Gu, Minheng Chen, Yue Yuan, Xin Guo, **Tian-Yi Zhou**, Qiang Fu.
Social Science & Medicine 364 (2025): 117553.
2. **Classification of Data Generated by Gaussian Mixture Models using Deep ReLU Networks.** (2024)
Tian-Yi Zhou, Xiaoming Huo.
Journal of Machine Learning Research (JMLR) 25(190): 1–54, 2024.
3. **Learning Ability of Interpolating Deep Convolutional Neural Network.** (2024)
Tian-Yi Zhou, Xiaoming Huo.
Applied and Computational Harmonic Analysis (ACHA) 68 (2024): 101582.
4. **Approximation and Non-parametric Estimation of Functions over High-dimensional Spheres via Deep ReLU Networks.** (2023)
Namjoon Suh, **Tian-Yi Zhou**, Xiaoming Huo.
International Conference on Learning Representations (ICLR) 2023.
5. **Gone with the Weed: Incidents of Adolescent Marijuana Use in the United States, 1976–2021.**

(2023)

Gu, Jiaxin, Xin Guo, Xiaoxi Liu, Yue Yuan, Yushu Zhu, Minheng Chen, **Tian-Yi Zhou**, Qiang Fu.
Annals of Epidemiology 88 (2023): 23-29.

6. Modified Poisson Regression Analysis of Grouped and Right-censored Counts. (2021)

Qiang Fu, **Tian-Yi Zhou**, Xin Guo.

Journal of the Royal Statistical Society: Series A 184(4): 1347-1367.

PREPRINTS

7. Improving Learning of Green's Functions Through Kernel Decomposition. (2026)

(α) Beiji Chen, Qiang Du, Kui Ren, **Tian-Yi Zhou**.

In Submission.

8. Adaptive continual learning by neural networks: Statistical insights for shared representation
(2026)

Tian-Yi Zhou, Qiang Du, Samory Kpotufe.

Preprint. (available upon request)

9. Synthetic Anomalies Improve Anomaly Detection: Learning Guarantees and Applications.
(2026)

Tian-Yi Zhou, Matthew Lau, Xiangchi Yuan, Jizhou Chen, Wenke Lee, Xiaoming Huo.

In Submission.

10. Learning to Detect Cyber Attacks: Neural Anomaly Detection for Cybersecurity with Theoretical Insights. (2026)

Tian-Yi Zhou*, Matthew Lau*, Jizhou Chen, Wenke Lee, Xiaoming Huo.

Minor Revision at Annals of Applied Statistics (AOAS).

11. Neural Operators for Nonlinear Functionals on RKHS. (2026)

Tian-Yi Zhou, Namjoon Suh, Guang Cheng, Xiaoming Huo.

Second-round Major Revision at Journal of Machine Learning Research (JMLR).

TEACHING EXPERIENCE

Guest Lecturer, APAM, Columbia University
APMA 6100: Masters' Research Seminar

Fall 2026

Guest Lecturer, APAM, Columbia University
APMA 4903/4901 Junior & Senior Seminar in Applied Mathematics

Fall 2026

Guest Lecturer, ISyE, Georgia Tech
ISyE 6740 Computational Data Analysis

Spring 2025

Graduate Teaching Assistant, ISyE, Georgia Tech
ISyE/CEE 3770 Statistics and Applications

Fall 2020, Spring 2021

Teaching Assistant, Department of Statistics and Actuarial Science,
University of Hong Kong
STAT 2601 Statistics and Probability I

Spring 2020

"Survival Cantonese" Tutor, Center of Development and Resources for Students,
University of Hong Kong
Conduct 1-to-1 Cantonese tutoring sessions to international students

2016-2017

HONORS AND AWARDS

Columbia University Data Science Institute Postdoctoral Fellowship

2025-2027

	Rising star in ISyE-MS&E-IOE	2025
🏆	Margaret and Stephen Kendrick Research Fellowship, Georgia Tech (Awarded annually to one Ph.D. student for research in machine learning and analytics)	2023-2024
	PhD mentoring service award, Georgia Tech	2023-2024
	Selected to Present at Cornell Young Researchers Workshop	2023
	Selected to present at Association for Women in Mathematics (AWM) Research Symposium	2023
🏆	Second Place in Student Paper Competition, The 13th American Institute of Mathematical Science (AIMS) Conference	2023
🏆	Selected Participant, FutureBAProf Workshop, Tippie College of Business, University of Iowa	2023
🏆	ARC-ACO Fellowship, Georgia Tech (Institute-wide scholarship to support research in high-impact cross-disciplinary data science)	2023
🏆	Runner-up, Alice and John Jarvis Ph.D. Student Paper Competition, Georgia Tech	2022
	Stewart Fellowship, Georgia Tech	2020-2021
	Overseas Research Fellowship, Columbia University (Supported by: Faculty of Science, University of Hong Kong)	2019
	Wong Shek Yung Student Enrichment Award, University of Hong Kong	2019
	Dean's Honors List, University of Hong Kong	2016-2017, 2017-2018
	Dean's Honors List, University of California, Los Angeles	Spring 2019
TRAVEL AWARDS	Travel Award for attending "Research at the Interface of Applied Mathematics and Machine Learning" conference (University of Houston)	2025
	Travel Award for attending "Statistics in the Age of AI" conference (George Washington University)	2024
	SIAM Travel Award (Special W. Randolph Fund Recipient) to attend SIAM PP24	2024
	The 13th AIMS Conference Travel Award (Supported by NSF)	2023
	Graduate Student Travel Award, College of Engineering and Student Government Association, Georgia Tech	2023
INVITED TALKS	INFORMS Annual Meeting 2026 Job Market Showcase Section Title: Statistical Learning, Markets, and Power Grids Time and Venue: November 1 (Sunday) 4:15 PM-5:30 PM, Moscone South-206 (Level 2) Learning operators on RKHSs SIAM Conference on Mathematics of Data Science (MDS26), Salt Lake City, UT November 2026 (scheduled) SIAM NY-NJ-PA Section Meeting, Rutgers University October 2026 (scheduled) Statistical learning theory of neural networks: from classification to anomaly detection in cyber-security The 9th International Workshop in Sequential Methodologies, American University June 2026	

(poster) "Research at the Interface of Applied Mathematics and Machine Learning" conference, University of Houston December 2025
 INFORMS Annual Meeting, Atlanta, GA October 2025
 Joint Statistical Meeting, Nashville, TN August 2025
 Seminar Talk at the Department of Mathematics and Statistics, Georgetown University March 2025

Approximation and Generalization of RKHS Functionals by Neural Networks

Seminar Talk at the Department of Mathematics, Hong Kong Baptist University December 2024
 Southeast Applied and Computational Mathematics Workshop, Georgia Tech April 2024

Optimal Unsupervised Anomaly Detection by Neural Networks: Theory and Practice in Cyber-security

Optimization and Machine Learning Seminar, University of Hong Kong December 2024
 Joint Meeting of the New Zealand, Australian and American Mathematical Societies, Auckland, New Zealand December 2024
 Seminar Talk at the Department of Mathematics, Universität Bayreuth, Bayreuth, Germany November 2024
 SIAM Seminar, Georgia Tech November 2024
 ML@GT Student Conference, Georgia Tech August 2024
 Statistics and Data Science Seminar, University of Sydney, Sydney, Australia July 2024
 ACTION AI Institute Annual Review and Knowledge Expo (Lightening talk & Poster), UC Santa Barbara June 2024

Classification of Unbounded Data generated by Gaussian Mixture Models via Deep ReLU Networks

INFORMS Annual Meeting, Seattle, WA October 2024
 (poster) "Statistics in the Age of AI" Conference, The George Washington University May 2024
 Analysis and Data Science Seminar, SUNY Albany April 2024
 SIAM Conference on Parallel Processing for Scientific Computing (PP24), Baltimore, MD March 2024
 SIAM Seminar, Georgia Tech October 2023
 Algorithms & Randomness Center (ARC) Special Colloquium, Georgia Tech September 2023

Learning Ability of Interpolating Deep Convolutional Neural Network

(poster) Cornell Young Researchers Workshop, Cornell University October 2023
 (poster) Georgia Statistics Day, Georgia Tech October 2023
 (poster) The 2023 Association for Women in Mathematics (AWM) Research Symposium, Clark Atlanta University September 2023
 10th International Congress on Industrial and Applied Mathematics (ICIAM 2023), Waseda University, Tokyo, Japan August 2023
 The 13th American Institute of Mathematical Science (AIMS) Conference, Wilmington, NC June 2023

PROFESSIONAL
SERVICE

Referee:

Journals: Journal of the American Statistical Association (JASA), Bernoulli, Electronic Journal of Statistics (EJS), INFORMS Journal on Computing (IJOC), Technometric, Journal of Complexity,

Journal of Approximation Theory
Conferences: NeurIPS, ICML, ICLR, AISTATS

Institutional Service:

Co-organisator, ISyE Student Seminar, Georgia Tech	2024-2025
Co-organisator, SIAM Seminar, Georgia Tech	2023
Secretary, SIAM Student Chapter, Georgia Tech	2023, 2024
Reviewer, President's Undergraduate Research Awards, Georgia Tech	2023, 2024
Co-organisator, "Machine Learning + Dynamics" Seminar, School of Mathematics, Georgia Tech	2023
Ph.D. Mentor, ISyE Ph.D. peer mentoring program, Georgia Tech	2022-2023, 2023-2024

Session Chair at Conferences:

SIAM Conference on Mathematics of Data Science (MDS26), Salt Lake City, UT	2026
INFORMS Annual Meeting, Atlanta, GA	2025

Outreach:

Volunteer, Georgia Science Olympiad State Tournament, Atlanta, GA	2023
Exam Writer and Grader, AP Statistics competition (for high school students), Georgia Tech	2023

REFERENCES

- Samory Kpotufe (Professor)
 Department of Statistics, Columbia University
 Email: samory@columbia.edu
- Qiang Du (Fu Foundation Professor of Applied Mathematics)
 Department of Applied Physics and Applied Mathematics, Columbia University
 Email: qd2125@columbia.edu
- Xiaoming Huo (A. Russell Chandler III Chair Professor)
 H. Milton Stewart School of Industrial and Systems Engineering, Georgia Tech
 Email: huo@gatech.edu